9 **(a)** Fig. 9.1 shows a diagram of a transverse wave.

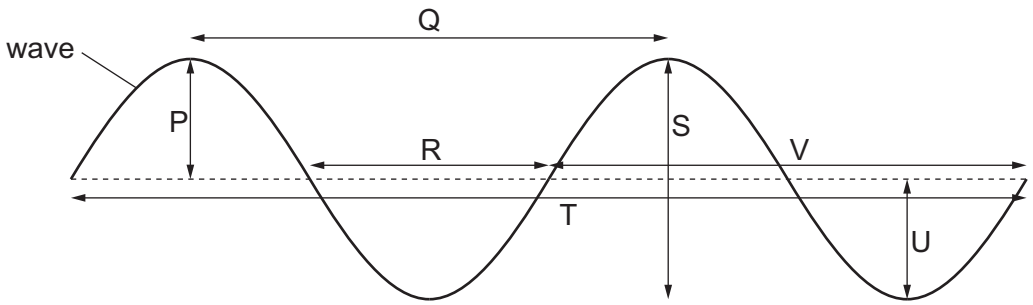

Fig. 9.1

From Fig. 9.1, identify all the lengths which represent **one** wavelength.

..... [1]

(b) Hydrogen in a very distant galaxy emits electromagnetic radiation which is observed on the Earth.

Scientists on the Earth measure the wavelength of the radiation from the very distant galaxy. The wavelength is 918 nm.

On the Earth, hydrogen in the laboratory emits electromagnetic radiation of wavelength 656 nm.

Name the effect that the scientists observe and state what this shows about the very distant galaxy.

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(c) Table 9.1 shows a wavelength of electromagnetic radiation from hydrogen observed in the laboratory and from three galaxies.

The galaxies are at different distances from the Earth.

Table 9.1

object	wavelength of hydrogen from object, observed on the Earth / nm
gas tube in laboratory	656
nearby galaxy	667
distant galaxy	750
very distant galaxy	918

Describe what Table 9.1 shows about the motions of the galaxies and state what this suggests is happening to the Universe.

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